Leakage-Aware Deep Learning for Financial Event-State Forecasting

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DL Course Project

Historical research artifact only: no trading advice, no live forecast, no positive benchmark claim.

Talk Roadmap

Question

- Can aligned company-news event states improve multi-horizon excess-return forecasts?
- Can deep models help without leaking future information?

Answer

- The strongest active evidence is a tuned XGBoost event-market baseline.
- A frozen-TSMixer mixer improved validation but not XGBoost lockbox.
- The main contribution is a rigorous, reproducible evaluation system.

Why This Is Hard

- Returns have low signal-to-noise and strong nonstationarity.
- News can matter, but timestamps, no-news days, and provider truncation create hidden biases.
- Model selection can overfit validation folds even when deep models look plausible.

Design principle

Train/validation is for search. Lockbox is for one final post-selection diagnostic.

Metric principle

ROC-AUC is not applicable: this is continuous excess-return regression.

Protocol: Leakage-Aware Artifact Pipeline

Stage	Artifact	Key guardrail	Output
Data	Top-50 + OHLCV + news	no raw overwrite	row universe
Features	market + event states	no-news rows preserved	sidecars
Splits	expanding folds + lockbox	purge/embargo	split manifest
Training	fold-local preprocessing	no lockbox tuning	saved preds
Evaluation	RMSE/MAE + rank IC	bootstrap + claim gate	reports

- Horizons: 1d, 5d, 10d, 20d, 60d.
- Lockbox ends 2026-02-13; all result roots keep manifests and prediction packets.

State of the Art Context

Financial ML baselines

- Efficient-market caution and expected-return multiple testing.
- Gradient boosting is still a very strong tabular baseline.
- Forecast metrics: RMSE, MAE, OOS R^2, rank IC.

Deep learning candidates

- FinBERT/BERT for semantic financial text features.
- TSMixer and Transformer-family sequence models.
- PatchTST, iTransformer, TFT, and Chronos motivate richer temporal architectures.

Implemented Model Stack

Text/Event

- Alpha Vantage aligned news.
- No-news as explicit state.
- FinBERT sidecars for later fusion.

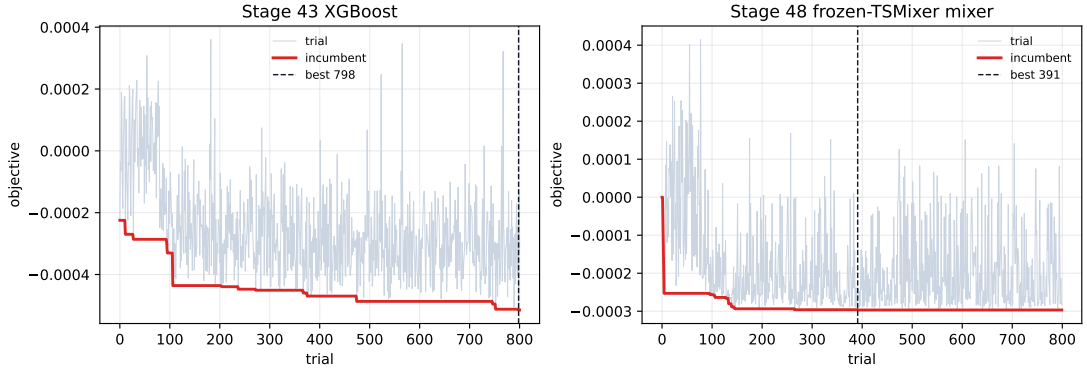
Market

- Lag/rolling features.
- Linear and fixed XGBoost baselines.
- Stage 43: 800-trial tuned XGBoost.

Sequence/Mixer

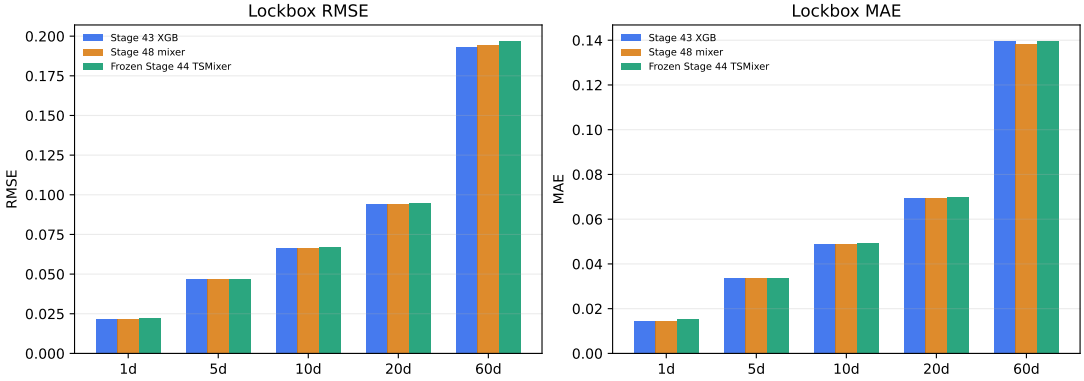
- TSMixer diagnostics.
- Readout correction audit.
- Stage 48: horizon mixer only.

Tuning Trace: Search Was Auditable



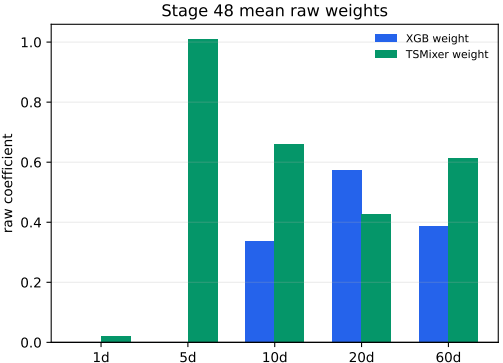
- Stage 43 XGBoost: 800 completed trials, best trial 798.
- Stage 48 mixer: 800 completed meta/mixer trials, best trial 391.
- Both searches used zero lockbox rows for tuning.

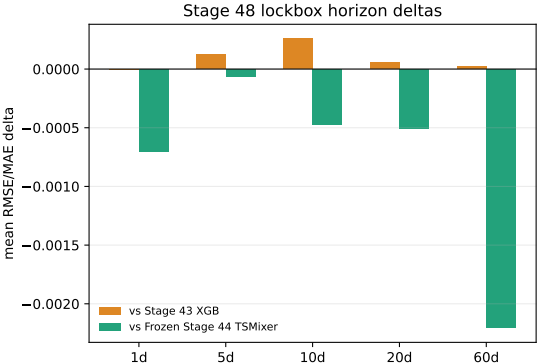
Primary Metrics: Lockbox RMSE and MAE



- Regression metrics are primary because labels are continuous forward excess returns.
- The Stage 48 mixer does not dominate Stage 43 XGBoost on lockbox.

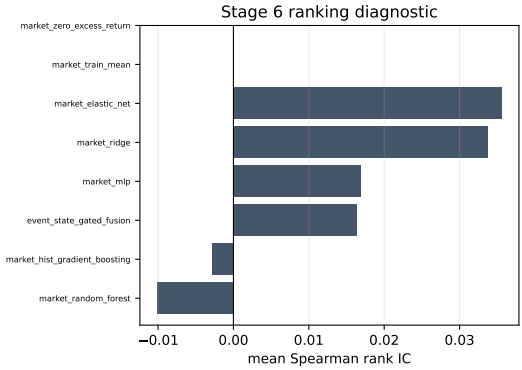
Deep Learning Lesson: Validation Gains Are Not Enough

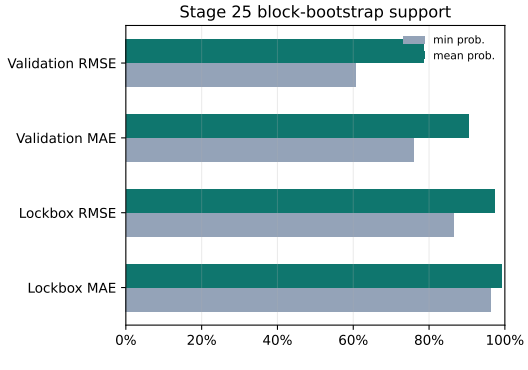




- Stage 48 improved validation weighted error vs XGBoost by -2.97×10^{-4}.
- It worsened post-selection lockbox weighted error vs XGBoost by $+9.20 \times 10^{-5}$.
- Conclusion: stronger architectures require stronger selection controls, not looser claims.

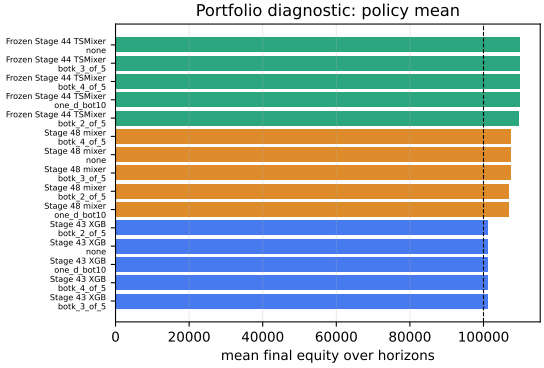
Secondary Evidence: Ranking and Uncertainty

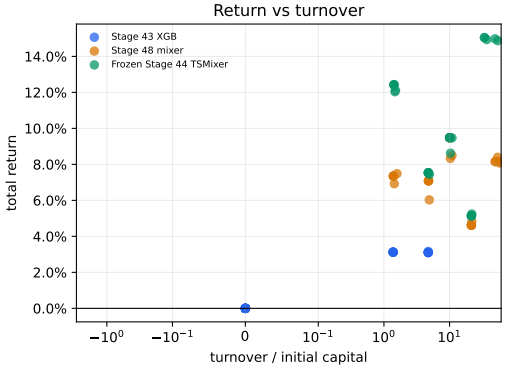




- Rank IC and hit-rate diagnostics are secondary: useful for interpretation, not primary selection.
- Moving-block bootstrap checks temporal dependence and uncertainty.

Portfolio Simulation: Diagnostic Only





- Initial capital: $100k; long-only, fractional shares, no costs, no slippage, no taxes.
- Best historical row: $115,045 final equity, but it uses frozen TSMixer diagnostic predictions.
- This is not a buy/sell system and not benchmark claim evidence.

Main Numerical Takeaways

Evidence	Result	Interpretation
Stage 43 XGB	lockbox delta -5.34×10^{-4}	active strongest evidence
Stage 43 XGB	62.2% both-metric improvement	non-claim improvement evidence
Stage 48 mixer	validation delta -2.97×10^{-4}	tuned signal looked useful
Stage 48 mixer	lockbox delta $+9.20 \times 10^{-5}$	did not beat XGBoost lockbox
ROC-AUC	not reported	target is continuous regression
Portfolio	diagnostic only	no trading/signal claim

Discussion: What I Learned

Positive lessons

- Reproducibility can be engineered through manifests, saved predictions, and tests.
- Tree baselines are hard to beat in small financial tabular settings.
- Deep sequence models need architecture audits and strong baselines.

Cautions

- Validation improvement did not guarantee lockbox improvement.
- Stale TSMixer readout invalidated earlier deep evidence.
- Portfolio-looking plots can mislead without transaction realism and causal validation.

Final Conclusion

Scientific conclusion

The updated XGBoost event-market pipeline is the current strongest artifact-backed evidence, but event-state value and deep-sequence superiority are not proven.

Deep learning conclusion

For financial forecasting, the most important DL competence is not just adding BERT, TSMixer, or Transformers; it is controlling leakage, validating architecture semantics, and resisting overfit narratives.

Non-claim boundary

No live forecast, no trading advice, no benchmark-comparability promotion, and no Stage 6 reinterpretation.

Speaker Script: 8 Minute Timing

Time	Script focus
0:00–0:40	State the problem: event-state forecasting is attractive but leakage-prone.
0:40–1:30	Explain the split protocol and why lockbox isolation is the core of the work.
1:30–2:30	Position SOTA: FinBERT/BERT, boosting, TFT/PatchTST/iTransformer/Chronos.
2:30–3:30	Explain the implemented model stack and the TSMixer correction.
3:30–4:40	Walk through tuning curves: 800 trials, no lockbox tuning.
4:40–5:50	Present RMSE/MAE and mixer lockbox degradation honestly.
5:50–6:40	Explain uncertainty/ranking and why AUC is not applicable.
6:40–7:20	Discuss portfolio diagnostics as non-claim evidence only.
7:20–8:00	Conclude with the methodological lesson and non-claim boundary.

Delivery Guidelines for the Grading Rubric

Presentation mechanics

- Logical sequence: problem, protocol, SOTA, models, results, discussion.
- Keep one claim per slide; point to figures rather than reading every number.
- Use 8 minutes, leaving 1–2 minutes for questions if the slot allows.

Delivery behavior

- Make eye contact after each slide transition.
- Stand still for key results; gesture only to highlight axes or deltas.
- Speak slowly on limitations: this demonstrates scientific maturity.

Likely Q&A

- **Why no AUC?** Labels are continuous excess returns; classification AUC would require an extra directional label and a different objective.
- **Why not claim the portfolio result?** It is historical, frictionless, post-selection, and partly uses stale/frozen TSMixer diagnostic predictions.
- **Where is deep learning?** FinBERT sidecars, TSMixer sequence modeling, horizon mixing, and comparison to modern time-series architectures; the project also shows why DL claims need robust validation.
- **What would improve next?** Corrected TSMixer rerun, stronger event encoders, larger universe as a new benchmark family, and cost-aware portfolio diagnostics.

Key References Mentioned

Fama (1970); Fama and French (1993); Campbell, Lo, and MacKinlay (1997); Tetlock (2007); Loughran and McDonald (2011); Devlin et al. (2019); Araci (2019); Chen and Guestrin (2016); Ke et al. (2017).

Lim et al. (2021); Nie et al. (2023); Liu et al. (2024); Ansari et al. (2024); Hyndman and Koehler (2006); Bergmeir et al. (2018); Harvey et al. (2016); Fawcett (2006).